

SAI MANJUNATH PATHAPADU

+91 6302348787 | saipathapadu@gmail.com | [linkedin.com/in/sai-manjunath-pathapadu](https://www.linkedin.com/in/sai-manjunath-pathapadu) | github.com/Manju-05

SUMMARY

Specializing in Machine Learning and Artificial Intelligence. Experienced in developing intelligent automation solutions across AI-powered chatbots and enterprise-grade workflow automation, with a strong focus on solving real-world challenges through AI and automation.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, SQL

AI / LLM Engineering: LangChain, RAG, ReAct Agents, Prompt Engineering & Caching, NLP

Machine Learning: Scikit-learn, Random Forest, SVM, Logistic Regression, Data Preprocessing, Exploratory Data Analysis

Web & Backend: React, Next.js, FastAPI, REST APIs, HTML5, CSS3

Data & Databases: DBMS, Supabase

Tools & Platforms: Git, GitHub, AWS Cloud Foundations, UiPath

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS

PROJECTS

CareerLens AI — AI-Powered Resume & Job Intelligence Platform

- Built an AI career platform using Hybrid RAG, RRF reranking, and ReAct agents to deliver grounded resume analysis, JD matching, ATS optimization, and skill-gap recommendations.
- Engineered the full stack with FastAPI, Next.js, LangChain, and ChromaDB/FAISS, supporting streaming LLM responses and scalable document retrieval.
- Designed interactive mock-interview flows and validated AI-generated outputs for accuracy before surfacing them to users.

TsunamiSense — Intelligent Seismic Risk Detection

- Engineered an AI-based tsunami risk classification framework using 1,334 USGS earthquake events, benchmarking Random Forest, SVM, and Logistic Regression models on classification metrics.
- Tuned a Random Forest model to 0.95 AUC and 0.92 PR-AUC, achieving strong discrimination between tsunami and non-tsunami events.

AgriAssist AI — AI-Powered Multimodal Agriculture Advisory System

- Engineered a low-latency multimodal AI system using Groq SDK, Llama-4 Scout Vision, and Redis for real-time crop recommendations, pest diagnosis, and soil analysis, achieving 91% diagnostic accuracy and cutting identification time from hours to seconds.
- Implemented prompt caching to reduce LLM token costs by 50%, and added multilingual voice/text interaction, enabling faster AI-assisted agricultural decision-making for farmers in field environments.

WORK EXPERIENCE

AgentRoom AI — Frontend Developer Intern *Oct 2025 – Jan 2026*

- Shipped a responsive production website and reusable UI components, collaborating with the team through Git-based workflows, Agile ceremonies, and code review.
- Debugged and optimized performance across features, incorporating peer feedback and communicating progress clearly to stakeholders.

ACHIEVEMENTS

- Competed in the Technoxian World Robotics Championship, Noida.
- Presented an AI-driven Scholarship Recommendation System at the Ratan Tata Innovation Hub (RTIH) RiseUp MSME Program.

CERTIFICATIONS

- Oracle Certified AI Foundations Associate — Oracle

EDUCATION

Rajeev Gandhi Memorial College of Engineering and Technology, Nandyal *Jul 2024 – Apr 2027*

B.Tech, CSE — Artificial Intelligence and Machine Learning (83%)

SVR Engineering College, Nandyal *Oct 2021 – Apr 2024*

Diploma in Computer Engineering (93%)